

Collection, Proccession& Presentation of Data

Data:

Data is the plural form of datum. In short, data is the numerical value of facts. **It is originally a latin noun meaning 'something given'.** **Data are** the function stones and basic raw material in relation to any statistical investigation. **Data** is a set of values of qualitative or quantitative variables. Data is measured, collected, reported and analyzed, whereupon it can be visualized using graphs, images or other analysis tools. Finally we can say that, **data is a collection of facts, such as numbers, words, measurements, observations or even just descriptions of things.**

Sources of data

On the basis of sources of data, data may be classified into primary data and secondary data:

Primary data: The data are collected through first hand investigation without any internal records or any published sources are called primary data. **Example:** survey data, interview data, questionnaire data etc.

Secondary Data: The data are collected from any medium which has already been collected by others are called secondary data. **Example:** Journal, thesis, published statistics etc.

Methods Used for Collecting Primary Data:

Primary data are those which are collected for the first time and so are in crude form. Primary data are always collected from the source. It is collected either by the investigator himself or through his agents. There are different methods of collecting primary data. Each method has its relative merits and demerits. The investigator has to choose a particular method to collect the information.

Following are given the methods for collection of such data:

- (1) Direct personal investigation.
- (2) Indirect oral investigation.
- (3) Through local correspondents.
- (4) Schedules sent through enumerators
- (5) By Email
- (6) Mailed Questionnaire.

(1) Direct personal Investigation:

Here, the investigator himself visits the persons those are source of the data and collects necessary information either through interview with the persons concerned or through observation of the data on the spot. This method is suitable where intensive study of any phenomenon is required.

This method has many merits and demerits which are given as following:

Merits:

1. The data is more reliable as they are obtained directly.
2. Sensitive questions can be avoided.
3. There is chance of more response in it.
4. The questions can be adjusted according to the standard of the sources.
5. There is uniformity in data.

Demerits:

1. It is not suitable where the field is very vast and wide.
2. It is very much expensive.
3. It needs a large number of enumerators.
4. It takes long time to collect the data from all the persons.
5. It is purely subjective in nature, therefore the success of investigation depends mostly intelligence, skill, tact, insight, honesty and other qualities of investigators.

Suitability:

This method is suitable when:

1. Area of investigation is limited.
2. A very high degree of accuracy is required.
3. The result of investigation is to be kept secret.
4. Area of investigation is homogeneous i.e. having same qualities.

Precautions:

1. Investigator should be trained and skilled.
2. Investigator should not have personal bias.
3. Investigator should be polite and questions should be simple.
4. Investigator must belong to the culture of the informant.

(2) Indirect Oral Investigation:

Here, the investigator collects the data indirectly by interviewing persons who are known to be close to the original persons or the incidence. This method is adopted when the original persons are not found or found to be reluctant to provide the required information. In this method a list of questions is prepared, and the witnesses are invited and made to answer the questions. The answers are recorded by the investigators.

This method has its merits and demerits also:

Merits:

1. It can cover a wide area.
2. It needs less time, energy and money.
3. Third party does not conceal the facts.
4. Intelligence, skill and tact of the investigator brings accuracy.
5. It is simple and convenient.

Demerits:

1. The information obtained from the other persons may not be reliable.
2. The third parties may be biased.
3. The witness chosen may not be proper for the cause or not expert in this field.

Suitability:

This method is suitable when:

1. The direct approach to sources does not exist.
2. The person cannot be relied upon
3. The area of investigation is large.
4. The information is needed, is kept secret from person.

Precautions:

1. Witnesses should not be biased about the subject.
2. Respondent or witness should have adequate knowledge about the person, whose information is to be received.
3. Number of witnesses should be large.
4. A cross check must be undertaken by investigator.

(3) Information through Local Correspondents:

In this method, the investigator does not work but appoints local agents or enumerators in different parts of the area. These agents or enumerators are asked to collect information and transmit it to the investigator. This method is often adopted by government, semi-government and bigger private institutions to collect the information for any purpose.

Following are the merits and demerits of this method:

Merits:

1. This method is cheap or economical.
2. There is economy of time and labour and money.
3. This method can be used where the area of inquiry is wide.
4. Quality of data is better as enumerators are trained.

Demerits:

1. Data are not reliable which are collected by this method.
2. If high degree of accuracy is required this method is not adopted.
3. The data are not original if collected by this method.
4. Collected data lack uniformity because they are collected by different correspondents and they differ in quality and taste.

Suitability:

This method is suitable when:

1. Rough and approximate estimates are required.
2. The information is wanted at regular intervals.
3. The field of inquiry is wide.

Precautions:

1. Correspondents must be trained.
2. No. of correspondents should not be small.
3. Correspondents should not enter their own views.
4. Correspondents should be within from the culture of the informants.

(4) Schedules Sent through Enumerators:

In this method, list of questions or schedules are sent to the informants through the enumerators. They read the questions to the informants and record their answers on the same schedules. At first, enumerator explains the aims and objectives of the enquiry and asks them for co-operation.

The difference between the mailed questionnaire method and this method is that in the former case the informants themselves record the answers while in this case, the enumerators record the information. This method is used by governments and research institutions like National Council of Applied Economic Research (NCAER), Federation of Indian Chamber of Commerce and Industries (FICCI) etc.

Like the other methods, this method also has merits and demerits:

Merits:

1. Information can be got even from uneducated persons.
2. This information is more reliable and correct.
3. It covers wide area.
4. It is unaffected by the personal bias of the investigators.
5. There are fewer chances of non-responses as enumerator's visits personally.

Demerits:

1. It is costly because enumerators have to be paid.
2. It is time consuming as every informant is visited.
3. It requires trained enumerators, which are not easily available.
4. The personal bias of enumerator may lead to wrong conclusions.
5. It can only be used by big organizations.

Suitability:

This method is suitable when:

1. Area to be covered is wide.
2. No extra question is needed to be asked.
3. Service of enumerators is cheap and easily available.

Precautions:

1. Enumerators should be trained.
2. He should be educated to the required subject.

(5) By Email

The growing popularity of the internet brought a major shift in Email .It is hard to find a quick and easy way of reporting answers through the internet.

Merits:

1. It is useful when the area of investigation is wide'
2. It saves time, labour and money.
- 3.As the information is collected from knowledgeable persons ,it is expected to be useful and reliable.
- 4.The inquiry could be more extensive and different aspects or the problems can also be studied properly.

Demerits:

1. The information collected may be subjective and is subject to the personal bias e of the persons or agencies from whom it is collected.
2. One has to be very careful about the selection of persons from whom the information is collected, because their knowledge and personal attitude towards the problem greatly affect the quality of data So one has to be very careful in dealing with such situations.

(6) Mailed Questionnaires:

An important method for the collection of data used for large area is that of questionnaires. In this method, a list of questions is prepared relating to the problem under investigation, is printed and then sent out to the informants through post. It is requested that it may be returned to the investigator properly filled up. A covering letter is also sent with the questionnaire. A stamped self addressed envelope is also attached. Following are the merits and demerits of this method:

Merits:

1. There is the economy of time, labour and wealth.
2. This method is used if the field of inquiry is wide.
3. There is uniformity in the data because informants are directly involved,

Demerits:

1. Information can be collected from the educated persons only.
2. Its response is uncertain.
3. Supplementary questions cannot be asked as questions are fixed.
4. If questionnaire is tough, it may not be responded.
5. There are chances of non-response due to indifference of attitude.

Suitability:

This method is suitable when:

- (i) The informant is educated.
- (ii) The area of coverage is very wide.

Precautions:

1. Questions should be in a good sequence.
2. Questions should not be personal, as informant will not want to answer.
3. A request letter must be attached.
4. It should be accompanied by a self addressed, stamped envelope to ensure reply.

Uses of Primary Data :The common uses of primary data are mentioned below:

- (1) They are used as the basis of measurements.
- (2) They are used for original study.
- (3) They are direct investigations.
- (4) They are used for very recent information.
- (5) They are used for reducing uncertainty.

Methods/Sources of Collecting Secondary Data

Secondary data can be collected by the following sources:

- (1) Journals,
- (2) Reports,
- (3) Institutional publications,
- (4) Public library,
- (5) Information bureau,
- (6) Govt. agencies,
- (7) International agencies (UNO, FAO, WHO etc),
- (8) National and international news media,
- (9) Statistical books, etc

Secondary data are second hand information. They are not collected from the source as the primary data. In other words, secondary data are those which have already been collected. So they may be relatively less accurate than the primary data. Secondary data are generally used when the time of enquiry is short and the accuracy of the enquiry can be compromised to some extent. Secondary data can be collected from a number of sources which can broadly be classified into two categories.

- i) Published sources
- ii) Unpublished sources

Published Sources:

Mostly secondary data are collected from published sources. Some important sources of published data are the following.

1. Published reports of Central and State Governments and local bodies.(BBS)
2. Statistical abstracts, census reports and other reports published by different ministries of the Government.
3. Official publications of the foreign Governments.
4. Reports and Publications of trade associations, chambers of commerce, financial institutions etc.
5. Journals, Magazines and periodicals.
6. Periodic Publications of Government organizations like Central Statistical Organization (C. S. O.), National Sample Survey Organization (NSSO).
7. Reports submitted by Economists, Research Scholars, Bureaus etc.
8. Published works of research institutions and Universities etc.

Unpublished Sources:

Statistical data can also be collected from various unpublished sources. Some of the important unpublished sources from which secondary data can be collected are:

1. The research works carried out by scholars, teachers and professionals.
2. The records maintained by private firms and business enterprises. They may not like to publish the information considering them as business secret.
3. Records and statistics maintained by various departments and offices of the Central and State Governments, Corporations, Undertakings etc.

Secondary data are already collected information. They might have been collected for some specific purposes. So they must be used with caution. It is generally very difficult to verify such information to find out inconsistencies, errors, omissions etc. Therefore scrutiny of secondary data is essential. Because the data might be inaccurate, unsuitable or inadequate. Thus it is very risky to use statistics collected by other people unless they have been thoroughly edited and found reliable, adequate and suitable for the purpose.

Uses of Secondary Data

The common uses of secondary data are mentioned below:

- (1) They are used for decisions making.
- (2) They are suitable for multipurpose uses.
- (3) They are used for reference purposes.
- (4) They are used for documentary project.
- (5) They are used for minimizing costs.

Differences between primary data and secondary data

Basis for Comparison	Primary Data	Secondary Data
Meaning	Primary data refers to the first hand data gathered by the researcher himself.	Secondary data means data collected by someone else earlier.
Data	Real time data	Past data
Process	Very involved	Quick and easy
Source	Surveys, observations, experiments, questionnaire, personal interview, etc.	Government publications, websites, books, journal articles, internal records etc.
Cost effectiveness	Expensive	Economical
Collection time	Long	Short
Specific	Always specific to the researcher's needs.	May or may not be specific to the researcher's need.
Available in	Crude form	Refined form
Accuracy and Reliability	More	Relatively less

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